Supplementary Material

Safety and Efficacy of Intermittent Hypoxia Conditioning as a New Rehabilitation/Secondary Prevention Strategy for Patients with Cardiovascular Diseases: A systematic Review and Meta-analysis

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1-Glazachev 2017	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High risk	The study design was a nonrandomized, controlled, before-and-after trial.
Allocation concealment (selection bias)	Low risk	"All participants were blinded to group allocation."
Blinding of participants and personnel (performance bias)	Low risk	"Participants in the CTRL group were enrolled in the study after completing a standard rehabilitation program lasting 8 weeks (16 sessions in total) and were exposed to sham-IHHT (normobaric normoxic air) following a protocol/schedule similar to the IHHT group (15 daily sham sessions over 3 weeks)." "During each session of both the IHHT and sham-IHHT treatments, all participants were continuously monitored (blinded) using fingertip pulse oximeter (pulse rate and SaO2) and supervised by physicians and/or nurses."
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Low risk	All pre-specified outcomes were adequately reported
Other bias	low risk	The study appears to be free of other sources of bias
2-Dudnik 2018	**Risk of bias**	**Author judgment**
Random sequence generation	Low risk	"Patients were randomly (sealed envelope) allocated to control (CTRL) or intervention (IHHT) groups."

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(selection bias)		
Allocation concealment (selection bias)	High risk	"Group allocation concealment was not possible because CTRL group participants were asked to exercise, while IHHT were asked to ''keep their physical activity habits'' and not to participate in any structured exercise program during the 5 weeks of IHHT exposure."
Blinding of participants and researchers (performance bias)	Unclear	Not mentioned
Blinding of outcome assessment (detection bias)	Unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low risk	The study appears to be free from other sources of bias
3-Tuter 2018	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	Low risk	"Five days before the operation, patients were randomly assigned to three groups using a computer-generated randomisation table: two main groups (IHHT and RIP) and a IHHT-control group."
Allocation concealment (selection bias)	Low risk	"Patients in the IHHT-control group also underwent four daily procedures before surgery using 40 min training periods with simulation of IHHT by using the same equipment, whereas moistened air was delivered through the mask"
Blinding of participants and personnel (performance bias)	unclear risk	"Only the person who conducted the training knew about the patient's allocation to a particular intervention group. Anaesthesiologists and cardiac surgeons had no access to this information."
Blinding of outcome assessment (detection bias)	Unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	low risk	The study appears to be free from other sources of bias
4-Bayer 2017	**Risk of bias**	**Author judgment**

Random sequence generation (selection bias)	Low risk	"It is a double-blind, randomized, stratified, and placebo-controlled study"
Allocation concealment (selection bias)	Low risk	"It is a double-blind, randomized, stratified, and placebo-controlled study"
Blinding of participants and personnel (performance bias)	Low risk	"The double-blinded study setting was ensured by the fact that neither patients nor therapists were aware about the type of breathing program."
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	low risk	The study appears to be free of other sources of bias
5-Burtscher 2004	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	Low risk	"The first 16 subjects meeting the inclusion criteria (8 with and 8 without prior myocardial infarction) comprised the study population."
Allocation concealment (selection bias)	Unclear	Not mentioned
Blinding of participants and personnel (performance bias)	Low risk	"study participants were randomly assigned in a double-blind fashion"
Blinding of outcome assessment (detection bias)	Unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	low risk	The study appears to be free from other sources of bias
6-Lymania 2011	**Risk of bias**	**Author judgment**
Random sequence generation	Unclear	Not mentioned

(selection bias)		
Allocation concealment (selection bias)	Unclear	Not mentioned
Blinding of participants and personnel (performance bias)	unclear risk	Not mentioned
Blinding of outcome assessment (detection bias)	Unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Low risk	-all pre-specified outcomes were adequately reported
Other bias	low risk	The study appears to be free from other sources of bias
7- Kalachev 2004	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High risk	The study design was a nonrandomized, controlled, before-and-after trial.
Allocation concealment (selection bias)	Low risk	Random allocation was done
Blinding of participants and personnel (performance bias)	Low risk	Mentioned
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low risk	The study appears to be free from other sources of bias
8- Makhova 2005	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High risk	The study design was a nonrandomized, controlled, before-and-after trial.
Allocation concealment (selection bias)	Low risk	Random allocation was done
Blinding of participants and personnel (performance bias)	Low risk	Mentioned
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low risk	The study appears to be free from other sources of bias
19- Rachok 2011	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High risk	The study design was a nonrandomized, controlled, before-and-after trial.

Allocation concealment (selection bias)	Low risk	Random allocation was done
Blinding of participants and personnel (performance bias)	Low risk	Mentioned
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low risk	The study appears to be free from other sources of bias
10- Ishchuk V.A. 2011	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High risk	The study design was a nonrandomized, controlled
Allocation concealment (selection bias)	Low risk	Random allocation was done
Blinding of participants and personnel (performance bias)	Low risk	Mentioned
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low risk	The study appears to be free from other sources of bias
11- Yelchainova 2005	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High risk	The study design was a nonrandomized, controlled
Allocation concealment (selection bias)	Low risk	Random allocation was done
Blinding of participants and personnel (performance bias)	Low risk	Mentioned
Blinding of outcome assessment (detection bias)	unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low risk	The study appears to be free from other sources of bias
12- Saeed 2012	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	High Risk	Non-randomized, non-control
Allocation concealment (selection bias)	High Risk	Non-randomized, non-control
Blinding of participants and personnel (performance bias)	Unclear	Not mentioned
Blinding of outcome assessment (detection bias)	Unclear	Not mentioned
Incomplete outcome data (attrition bias)	Low Risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low Risk	The study appears to be free from other sources of bias
13- Burtscher 2009	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	Low Risk	Randomization
Allocation concealment (selection bias)	Low Risk	Random allocation
Blinding of participants and personnel (performance bias)	Low Risk	Double-blind

Blinding of outcome assessment (detection bias)	Low Risk	Double-blind
Incomplete outcome data (attrition bias)	Low Risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low Risk	The study appears to be free from other sources of bias
14- Shatlio 2008	**Risk of bias**	**Author judgment**
Random sequence generation (selection bias)	**Unclear**	**Not mentioned**
Allocation concealment (selection bias)	**Unclear**	**Not mentioned**
Blinding of participants and personnel (performance bias)	**Unclear**	**Not mentioned**
Blinding of outcome assessment (detection bias)	**Unclear**	**Not mentioned**
Incomplete outcome data (attrition bias)	Low Risk	There was no drop outs during the follow up period
Selective reporting (reporting bias)	Unclear	-No adverse events were reported
Other bias	Low Risk	The study appears to be free from other sources of bias